

Boundary-Aware Training for Dental Radiograph Instance Segmentation

Technical report

Abstract

We evaluate boundary-aware and class-balanced training objectives for instance segmentation on a 31-class panoramic dental radiograph corpus, and report that none improves on a tuned stock baseline once per-class contributions are decomposed. The result is not a null for want of trying: seven ablation cells at matched budget, two backbones, three architectural changes to the mask head, and a downstream clinical endpoint all land within seed-level spread. Four things make the null informative. The corpus as distributed cannot support evaluation at all, and we supply a verified-disjoint replacement. We measure the ceiling the mask representation imposes, independent of any model, and show that ceiling is *not* the binding constraint—which correctly predicts the failure of a higher-resolution mask head. A closed-form oracle fit then attributes 85 % of the mask deficit to coefficient prediction rather than representation. And acting on that attribution fails in both available directions: tripling the coefficient head’s capacity leaves masks unchanged while improving the detector by accident, and supervising the coefficients directly against their closed-form optimum makes masks worse. The deficit is correctly localised and not thereby actionable. What *was* actionable lay upstream of the network entirely: forty percent of the corpus is natively 1615×840 and was being trained at 640. Restoring the input resolution improves segm mAP by **+1.50 pp (+14.2 % relative)** and strict-IoU AP75 by +47 %, with the gain concentrated in the small clinical findings (caries +35 %, periapical lesion +29 %), and it is the only intervention in this study that survives every check the apparatus can mount.

1 The Corpus Cannot Be Evaluated as Distributed

The dataset ships a train/valid/test split which is unusable: 1,516 of 1,580 test files have their source image in train or valid (95.9 %), only 64 test sources are genuinely unseen, and 194 patients appear on both sides. Augmented copies were distributed across partitions after augmentation rather than before.

Two further defects: a spurious `croen` category at id 12 in the test COCO file shifts every higher class id out of alignment, and `data.yaml` points `test:` at the validation images.

We rebuild patient-grouped and class-stratified (9752 / 2090 / 2090; test coverage 13 $\rightarrow$ 29 of 31 classes) and verify disjointness twice by independent implementations: 0 exact duplicates, 0 NCC-confirmed near-duplicates across 792,811 candidate pairs, 0 shared patients. Maximum NCC is 0.9334 against a 0.98 threshold.

Perceptual hashing alone is not a leakage test on this modality. Every panoramic radiograph is the same anatomy in the same framing, so unrelated studies routinely fall within 5 bits on an 8×8 dHash. Without pixel confirmation, dHash reports all 792,811 pairs as leaks on a split that is in fact clean—a false-positive rate near 100 %.

Statistical power. Sixteen of 31 classes appear in fewer than 10 validation images; two appear in none; eleven tail classes score exactly zero. No long-tail claim is supportable and none is made.

2 The Objective

Let $M : \Omega \rightarrow \{0, 1\}$ be a ground-truth instance mask, $\hat{M}$ the predicted logits, $p = \sigma(\hat{M})$. For odd band width k with $S_k(i)$ the $k \times k$ window at pixel i :

$$\delta_k(x)[i] = \max_{j \in S_k(i)} x[j], \quad (1)$$

$$\varepsilon_k(x)[i] = \min_{j \in S_k(i)} x[j], \quad (2)$$

$$B_k(x) = \delta_k(x) - \varepsilon_k(x). \quad (3)$$

B_k is the morphological gradient, computed as one max-pool using $\varepsilon_k(x) = -\delta_k(-x)$. The objective is Dice between the two band maps rather than between regions:

$$\mathcal{L}_{\text{band}} = 1 - \frac{2 \sum_{\Omega} B_k(p) \odot B_k(M) + \epsilon}{\sum_{\Omega} B_k(p) + \sum_{\Omega} B_k(M) + \epsilon}, \quad (4)$$

added to the stock cropped BCE with weight λ . Throughout $k = 3$, $\lambda = 0.5$, never swept.

Mechanism. $\partial \delta_k / \partial x$ is nonzero only at each pooling window’s arg-max. In a confidently predicted interior the window is locally constant, so $\delta_k = \varepsilon_k$ and no gradient flows. The term acts only on pixels at or beside the predicted edge—measured at **48.5 % nonzero gradient versus 100 % for region losses**. The predicted signature is therefore AP75 up, AP50 down, not a uniform lift.

Novelty, stated honestly. The construction is a synthesis of known components: morphological gradient as a differentiable edge extractor, soft Dice [2] as the overlap objective. Its nearest relative, Boundary IoU [1], is an evaluation metric. It is not an existing named loss applied unchanged, and it is not a new mathematical object.

3 Ablation and Test

arm	configuration	mAP	AP50	AP75	head
S0	reference	0.1055	0.2549	0.0661	0.2815
S1c	inverse-sqrt	0.1065	0.2591	0.0659	0.2836
S2	inv-sqrt + boundary	0.1050	0.2467	0.0730	0.2921
S3	inv-sqrt + copy-paste	0.1042	0.2570	0.0635	0.2836
S4	all three	0.1051	0.2513	0.0675	0.2916

Table 1: Validation, 50 epochs, mask metrics. The grid is cumulative, so $S2 - S0$ conflates weighting with the boundary term. The valid isolation is $S2 - S1c$: AP75 +0.70 pp (+10.7 %), head +0.85 pp (+3.0 %), AP50 −1.23 pp.

Class weighting does nothing, mechanically. YOLO’s task-aligned assigner selects positives by $\text{cls}^\alpha \times \text{IoU}^\beta$. For a class the model never learned, $\text{cls} \approx 0$, so the ground truth receives few positive assignments and re-weighting scales a loss over assignments never made. The bottleneck is upstream of the loss.

Copy-paste hurts, and the mechanism generalises. It added $\sim 10\times$ more tail instances and worsened every metric that matters. In panoramic radiographs anatomical position is part of the class definition, so a finding pasted into an anatomically impossible location teaches the model to detect the paste rather than the pathology.

model (100 epochs)	mAP	AP50	AP75	tail
baseline	0.1051	0.2590	0.0687	0.0365
boundary alone	0.1071	0.2585	0.0680	0.0441
weighting + boundary	0.1007	0.2430	0.0650	0.0261
weighting alone	0.0973	0.2369	0.0612	0.0252

Table 2: Held-out test. The nominal +0.21 pp of the boundary arm decomposes to three classes with 4, 8 and 8 test instances. AP50, AP75 and head are flat.

Validation ordering did not transfer, three times. S1c led on validation and came last on test. The +10.7 % AP75 gain at 50 epochs on validation is -0.37 pp at 100 epochs on test.

4 The Representation Ceiling Is Not the Constraint

Round-tripping every ground-truth mask through the prototype grid bounds what any model could achieve. At the stock 160×160 grid, mean Dice is 0.8963 and **17.7 % of instances cannot reach IoU 0.75**; root canal treatment is capped at 0.622 and caries at 0.721, both below threshold.

But the model achieves Dice **0.6969**—only 78 % of the ceiling it already has. Doubling the grid raises a limit nothing is pressing against.

We tested it: a prototype head fed from P2 (stride 4) rather than P3 (stride 8), using sub-pixel convolution [3], with **878,432 fewer parameters**. Against a matched control at identical budget: -0.72 pp mAP, -0.49 pp AP75. The prediction was wrong and the experiment says so.

5 Metric Failure: Distances on Per-Model Subsets

HD95 and ASSD are undefined on empty masks, so they average over a denominator that differs per model. Averaged per model, the boundary objective appeared **17.3 % better on HD95 and 24.6 % better on ASSD with disjoint intervals**. Recomputed on the 5,385 cases both arms scored, paired and bootstrapped over images, HD95 is +2.30 *worse* with the interval excluding zero, and every other metric is inseparable from zero. Correcting the denominator inverted the sign.

A second failure has the opposite shape and is reported in Section 8: an arm gains +0.64 pp of segmentation AP, the largest gain in this study, while its masks do not improve at all. Segmentation AP is a conjunction—a correct detection *and* a correct mask—so a change that helps only the detector raises it without touching mask quality. Scoring the same predictions as boxes separates the two, and here the box gain is the larger of the pair. Neither failure is exotic; both are the default reading of a standard metric.

6 Clinical Endpoint

Endpoint: bone-loss area as a percentage of image area. Distance-based and normalized bone-level endpoints are *not computable*—both require cemento-enamel-junction and root-apex landmarks, and the annotations provide a region polygon only; the images carry no pixel-spacing metadata.

Paired per-image absolute error: +0.0154 pp, 95 % CI $[-0.0738, +0.1060]$. No detectable change. The arms differ in operating point, not accuracy: the boundary term carries a larger negative bias while producing roughly half the false-positive area on healthy radiographs.

7 The Mask Deficit Is Coefficient Prediction, Not Representation

The 20-point gap between what the model achieves (Dice 0.6969) and what its prototype grid allows (0.8963) had no attribution. Prototype-based heads reconstruct each mask as $\sigma(\mathbf{c}^\top P)$, so either the learned 32-dimensional basis cannot represent these shapes, or the basis is adequate and the coefficient head fails to locate the right point in it. Those call for opposite fixes.

We separate them without training. For every ground-truth instance we solve, in closed form, for the coefficient vector that best reconstructs it from the model’s own prototypes,

$$\mathbf{c}^* = \arg \min_{\mathbf{c}} \|P^\top \mathbf{c} - y\|^2, \quad (5)$$

over the instance’s box crop, where $y \in \{-1, +1\}$ is the ground-truth mask. Thresholding $P^\top \mathbf{c}^*$ gives the best mask *any* coefficient head could produce from that basis.

quantity	Dice
grid ceiling (what the resolution allows)	0.8963
oracle coefficients, full resolution	0.8659
oracle coefficients, at prototype resolution	0.9636
what the trained model achieves	0.6969

source of the 0.1994 gap	Dice lost	share
prototype basis cannot represent it	0.0304	15 %
coefficient head fails to find it	0.1690	85 %

Table 3: 4123 instances. The basis is not the constraint: handed optimal coefficients the model jumps from 0.70 to 0.87.

This closes the loop on every negative result above. Boundary terms and comparator losses reshape the *objective*; higher prototype resolution enlarges the *representation*. Neither is the binding constraint, which is why neither moved the number, and why the P2-fed head performed worse rather than better—it added resolution to a basis that was never the limit.

The head capacities make this concrete. In YOLOv8x-seg the coefficient branch bottlenecks 320 to 80 channels and carries 1.33 M parameters, against 2.26 M for the prototype generator and 7.41 M for classification. The smallest of the three heads is responsible for 85 % of the mask deficit.

8 Acting on the Attribution: Both Routes Fail

An attribution is only useful if it is actionable, so we tested the two things it implies. If the coefficient head is the constraint, then either it lacks the *capacity* to express the mapping, or it lacks a training *signal* that reaches it. Every objective tried earlier supervises in pixel space, where the gradient must travel back through the prototype product to reach the coefficients at all.

Capacity (K2). The coefficient branch `cv4` is rebuilt with its hidden width raised from 80 to 256 channels, $0.29 \rightarrow 3.86$ M parameters in the branch. Nothing else changes.

Supervision (K1b). An auxiliary term supervises the coefficients directly against the closed-form optimum $\mathbf{c}^*$ of the previous section, recomputed each step from the model’s own prototypes and detached, so the basis cannot move to meet the prediction:

$$\mathcal{L}_{\text{coeff}} = \frac{\|\mathbf{c}_{\text{pred}} - \mathbf{c}^*\|^2}{\|\mathbf{c}^*\|^2 + \varepsilon}. \quad (6)$$

A numerical caution that is part of the result. The first implementation used a fixed ridge $\lambda = 10^{-3}$ in the solve for $\mathbf{c}^*$. The normal-equation matrix $A = PBP^\top$ is built from learned features restricted to one instance box, and on COCO-pretrained prototypes over real instances its eigenvalues span 10^{-6} to 10^3 , with the smallest slightly negative from rounding. A fixed ridge regularises some instances and leaves others singular; one singular instance out of 95,745 annotations is enough. Training loss ran 1.8×10^5 at epoch 1, NaN by epoch 10 and 7.2×10^7 by epoch 40, against 2.9 falling to 1.6 without the term. The arm still produced a plausible-looking mAP of 0.1106, which measures the defect and not the method. The weight had been calibrated on the *converged* reference model, where $\overline{\mathbf{c}^{*2}} = 2.86$; on a run initialised from COCO weights the same quantity measures 4 to 53 with a far heavier tail. Making the ridge relative to $\text{tr}(A)/n$ bounds the condition number by construction and removes the failure; the results below use that form.

arm	segm mAP	Δ	box mAP	Δ	tail
S0 reference	0.1055	—	0.1551	—	0.0101
K1b supervision	0.0985	−0.70	0.1542	−0.09	0.0039
K2 capacity	0.1118	+0.64	0.1625	+0.74	0.0297

Table 4: Validation, 50 epochs, seed 42, otherwise identical to the reference. Deltas in percentage points. K2 is the largest gain of any arm in this study and exceeds the ± 0.21 pp noise floor threefold.

K2’s gain is not a mask gain. Scoring the *same* predictions as boxes rather than masks gives +0.74 pp, *larger* than the +0.64 pp segmentation gain. *cv4* emits mask coefficients and has no path to the box head, so the improvement cannot originate there: rebuilding the branch with three times the parameters perturbs the gradients returning into the shared neck, and the detector, not the mask, is what improves. Because segmentation AP requires a correct detection *and* a correct mask, better detection lifts it even when masks are unchanged. Paired on the 5352 cases where both models emit a mask (Table 5) the masks are flat to worse, and the one separable metric is negative.

metric	S0	K2	difference (95 % CI)
Dice	0.7043	0.7028	−0.0016 [−0.0043, +0.0013]
IoU	0.5784	0.5770	−0.0014 [−0.0041, +0.0019]
boundary F	0.7971	0.7928	−0.0043 [−0.0077, −0.0007]
HD95 (px)	64.15	63.95	−0.20 [−2.65, +2.11]
ASSD (px)	17.80	17.62	−0.18 [−0.88, +0.49]

Table 5: Paired on the same 5352 cases, 500-resample image-level bootstrap. Only boundary F is separable from zero, and it favours the *reference*. Read against Table 4, an arm can gain 0.64 pp of segmentation AP while its masks get no better.

K1b fails on its merits. With the target numerically sound the arm is a fair test, and it is negative: −0.70 pp, paired Dice −0.0017 and not separable. The mechanism is visible in the loss. At epoch 50

K1b’s total segmentation loss is 5.05 against the reference’s 1.65; the auxiliary term is capped at 2.0 by its clip, so K1b’s own pixel BCE is roughly 3.05 against 1.65. The term did not help the head find the pixel optimum, it pulled the head away from it. A scale mismatch would be the obvious explanation and is not the correct one: the oracle target and the head’s own coefficients agree to within 3 % in RMS (0.473 against 0.460). The coefficients that minimise pixel BCE and those that best reconstruct the mask in least squares are simply different points, and moving toward the second costs the first.

Consequence. The attribution is a correct *localisation* and not an actionable one. The coefficient head is where the error lives, and it is short of neither parameters nor a direct training signal. Adding capacity leaves mask quality unchanged and improves the detector by accident; adding coefficient-space supervision makes masks worse.

9 The Constraint Was the Input

With the coefficient head exonerated of capacity and signal failures, the remaining untested variable was what the network is shown. Forty percent of the corpus is natively 1615×840 ; at the training size of 640 its long side is reduced $2.5\times$, and the classes at the metric floor (caries, periapical lesion, bone loss) are precisely the small, low-contrast findings such a reduction starves. Fine-tuning the reference model at 1280 for 25 epochs, all else identical:

metric	baseline (640)	1280	change
segm mAP	0.1055	0.1204	+1.50 pp / +14.2 %
AP50	0.2549	0.2724	+6.9 %
AP75	0.0661	0.0974	+47.4 %
box mAP	0.1551	0.1568	+0.17 pp, inside the ± 0.75 pp floor

Table 6: The resolution arm against the reproduced baseline on the clean validation split.

Three checks, each of which eliminated an earlier candidate. The box control: box mAP moves +0.17 pp against its own measured ± 0.75 pp seed floor, so unlike the widened-head arm the segmentation gain has nowhere to originate except mask quality. The strictness split: AP75 rises seven times harder than AP50, the signature of better localisation rather than of ranking or confidence artefacts, which move both together. The paired test: on the 5,303 cases where both models emit a mask, IoU is separably better (+0.0048, CI [+0.0015, +0.0082]), the first arm in this study to achieve that; boundary F moves the other way (−0.0096), so the masks overlap the truth better while placing their contours less smoothly, which is why the gain expresses in AP75 rather than in boundary fidelity.

Where the gain lands. Root canal treatment +40 %, caries +35 %, periapical lesion +29 %, root piece +20 %, implant +22 %; tail-group AP roughly triples (0.0101 $\rightarrow$ 0.0295). Bone loss moves −25 %, the honest exception, and an informative one: it is a large diffuse region, not a small finding, so the mechanism predicts no benefit there and none appears.

The earlier resolution arms, reinterpreted. XP2 and XP3 raised the *prototype* resolution at 640 input and were the worst arms in the study (−2.02 and −1.30 pp). Both experiments are the same lesson from opposite sides: representation cannot exceed the information the input carries, and once the input carries it, the representation is no longer the limit.

Status of the claim. Two confounds are open and both are being closed by queued runs. The arm fine-tunes from the converged baseline, so it has seen 75 epochs against the baseline’s 50; the mitigating evidence is that additional epochs *hurt* everywhere else in this study, the baseline itself peaking at epoch 26 and losing 1.68 pp by epoch 50. And it is a single seed; two from-scratch replicates at independent seeds, trained for *fewer* epochs than the baseline, are the closing test. Externally, the gain currently trades sensitivity for precision: zero-shot on a professionally annotated corpus the 1280 model is more precise at tooth level (82.4 % against 73.6 %) but recalls less (32.5 % against 37.9 %), and inference at a scale-matched larger size worsened rather than recovered it, so the specialisation is real and the identified remedy is multi-scale training rather than inference-time rescaling.

10 Noise Floor (Measured)

The reference configuration was trained at three seeds, everything else identical. Determinism is exact on this pipeline, so seed choice is provably the only source of variation.

seed	mAP	AP50	AP75	head	tail
42	0.1055	0.2549	0.0661	0.2815	0.0101
1337	0.1056	0.2573	0.0679	0.2837	0.0117
2024	0.1037	0.2568	0.0629	0.2823	0.0105
sd	0.0010	0.0013	0.0025	0.0011	0.0008

Table 7: Noise floor at 2 sd: mAP ± 0.21 , AP50 ± 0.26 , AP75 ± 0.50 , head ± 0.22 , tail ± 0.17 pp.

Three consequences, and they are not the same verdict. The seven-arm ablation spans a 0.36 pp range, *inside* the band—selecting a winner on mAP was selecting noise. The best segmentation arm on test is +0.21 pp, sitting *exactly* at the threshold. But the isolated boundary effect on AP75 is +0.70 pp against a ± 0.50 pp floor, so it **exceeds** noise on validation at 50 epochs, and then reverses on test at 100 epochs. That is a **transfer failure, not a noise result**: the objective does what its mechanism predicts under the conditions it was measured in, and that does not survive a change of split and schedule.

11 Conclusion and Limitations

On this corpus, boundary-aware and class-balanced objectives do not improve instance segmentation beyond seed-level variation, and neither does acting on the one place the deficit was traced to. What the work establishes instead is that the distributed split cannot support evaluation; that perceptual hashing alone is not a leakage test here; that the representation ceiling is real but not binding; that the mask deficit is 85 % coefficient prediction, which is a correct localisation and an unactionable one; that distance metrics on per-model subsets invert their sign and segmentation AP can rise on a purely detection-side change; and that anatomical position is part of the class definition in radiographs.

The two arms of Section 8 are the sharpest statement of the negative result. Handed optimal coefficients the model jumps from Dice 0.70 to 0.87, so the head is demonstrably the constraint; given three times the parameters it produces the same masks, and given the optimal coefficients as an explicit target it produces worse ones. Whatever limits this head is not capacity and not the absence of a signal pointing at the right answer. The sequel is Section 9: the constraint was never inside the network. The one intervention that survives every check does not touch the objective, the sampling or the head; it restores information the pipeline was discarding before the network ever saw the image.

The published comparator losses did not run. Soft Dice, Tversky, Focal Tversky and Kervadec’s boundary loss are implemented behind a common interface, but the Kervadec arm failed with an in-place autograd error and the other three were queued behind it. The objective is therefore compared against the stock BCE baseline only, which is a necessary control and not a sufficient one.

Single seed on all reported arms, with the three-seed noise floor measured above. Single corpus, single backbone family. No prior published results exist for this dataset. No long-tail claim, because the split cannot support one.

References

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